

Given: Due:	Further Maths (Decision) HW 21				Name:	
Chapter 6: Linear programming						
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.						
91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)	
<i>Useful resources:</i> Your notes from recent lessons & your text book <u>Although most Complex arithmetic can be done on your calculator, show all steps to reach the answer & use calculator for checking results only.</u>						
Questions - do these on lined paper						
Q1. A manufacturer of frozen yoghurt is going to exhibit at a trade fair. He will take two types of frozen yoghurt, Banana Blast and Strawberry Scream. He will take a total of at least 1000 litres of yoghurt. He wants at least 25% of the yoghurt to be Banana Blast. He also wants there to be at most half as much Banana Blast as Strawberry Scream. Each litre of Banana Blast costs £3 to produce and each litre of Strawberry Scream costs £2 to produce. The manufacturer wants to minimise his costs. Let x represent the number of litres of Banana Blast and y represent the number of litres of Strawberry Scream. Formulate this as a linear programming problem, stating the objective and listing the constraints as simplified inequalities with integer coefficients. You should not attempt to solve the problem. (Total 6 marks)						
Q2. Class 8B has decided to sell apples and bananas at morning break this week to raise money for charity. The profit on each apple is 20p, the profit on each banana is 15p. They have done some market research and formed the following constraints. • They will sell at most 800 items of fruit during the week. • They will sell at least twice as many apples as bananas. • They will sell between 50 and 100 bananas. Assuming they will sell all their fruit, formulate the above information as a linear programming problem, letting a represent the number of apples they sell and b represent the number of bananas they sell. Write your constraints as inequalities. (7)						

Q3.

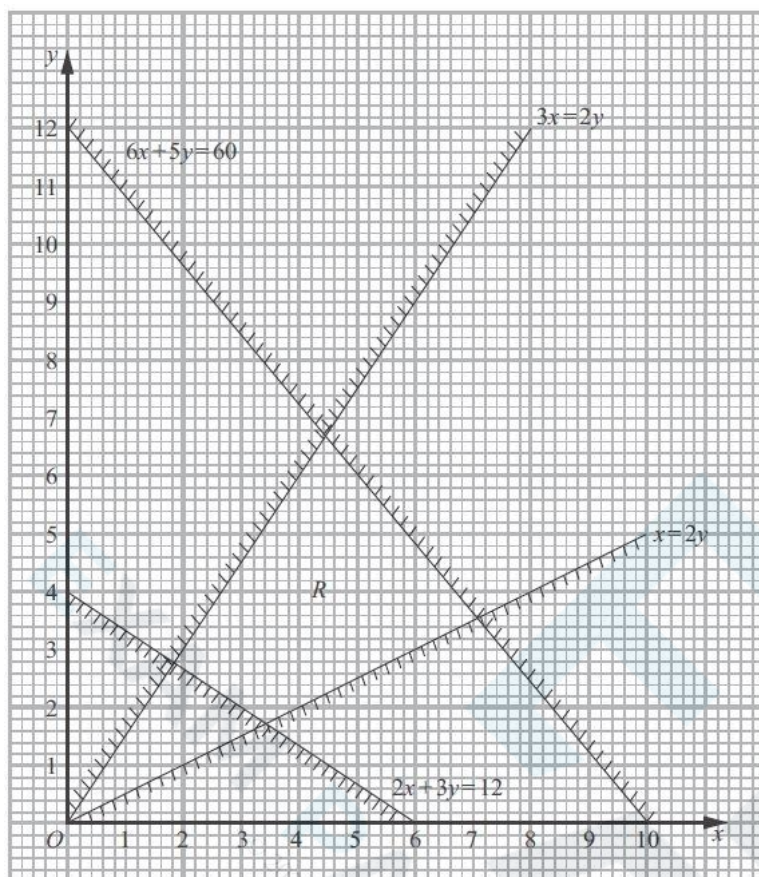


Figure 2

Figure 2 shows the constraints of a linear programming problem in x and y , where R is the feasible region.

(a) Write down the inequalities that form region R .

(2)

The objective is to maximise $3x + y$.

(b) Find the optimal values of x and y . You must make your method clear.

(4)

(c) Obtain the optimal value of the objective function.

(1)

Given that integer values of x and y are now required,

(d) write down the optimal values of x and y .

(1)



Q4. A linear programming problem in x and y is described as follows.

Maximise $P = 2x + 3y$

subject to

$$x \geq 25$$

$$y \geq 25$$

$$7x + 8y \leq 840$$

$$4y \leq 5x$$

$$5y \geq 3x$$

$$x, y \geq 0$$

(a) Add lines and shading to Diagram 1 in the answer book to represent these constraints. Hence determine the feasible region and label it R.

(4)

(b) Use the objective line method to find the optimal vertex, V, of the feasible region. You must clearly draw and label your objective line and the vertex V.

(3)

(c) Calculate the exact coordinates of vertex V.

(2)

Given that an integer solution is required,

(d) determine the optimal solution with integer coordinates. You must make your method clear.

(2)

Q5.

Ben is a wedding planner. He needs to order flowers for the weddings that are taking place next month. The three types of flower he needs to order are roses, hydrangeas and peonies.

Based on his experience, Ben forms the following constraints on the number of each type of flower he will need to order.

- At least three-fifths of all the flowers must be roses.
- For every 2 hydrangeas there must be at most 3 peonies.
- The total number of flowers must be exactly 1000

The cost of each rose is £1, the cost of each hydrangea is £5 and the cost of each peony is £4

Ben wants to minimise the cost of the flowers.

Let x represent the number of roses, let y represent the number of hydrangeas and let z represent the number of peonies that he will order.

(a) Formulate this as a linear programming problem in x and y only, stating the objective function and listing the constraints as simplified inequalities with integer coefficients.

(7)

Ben decides to order the minimum number of roses that satisfy his constraints.

- (b) (i) Calculate the number of each type of flower that he will order to minimise the cost of the flowers.
(ii) Calculate the corresponding total cost of this order.

(3)

END OF HOMEWORK

Total = 42 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.

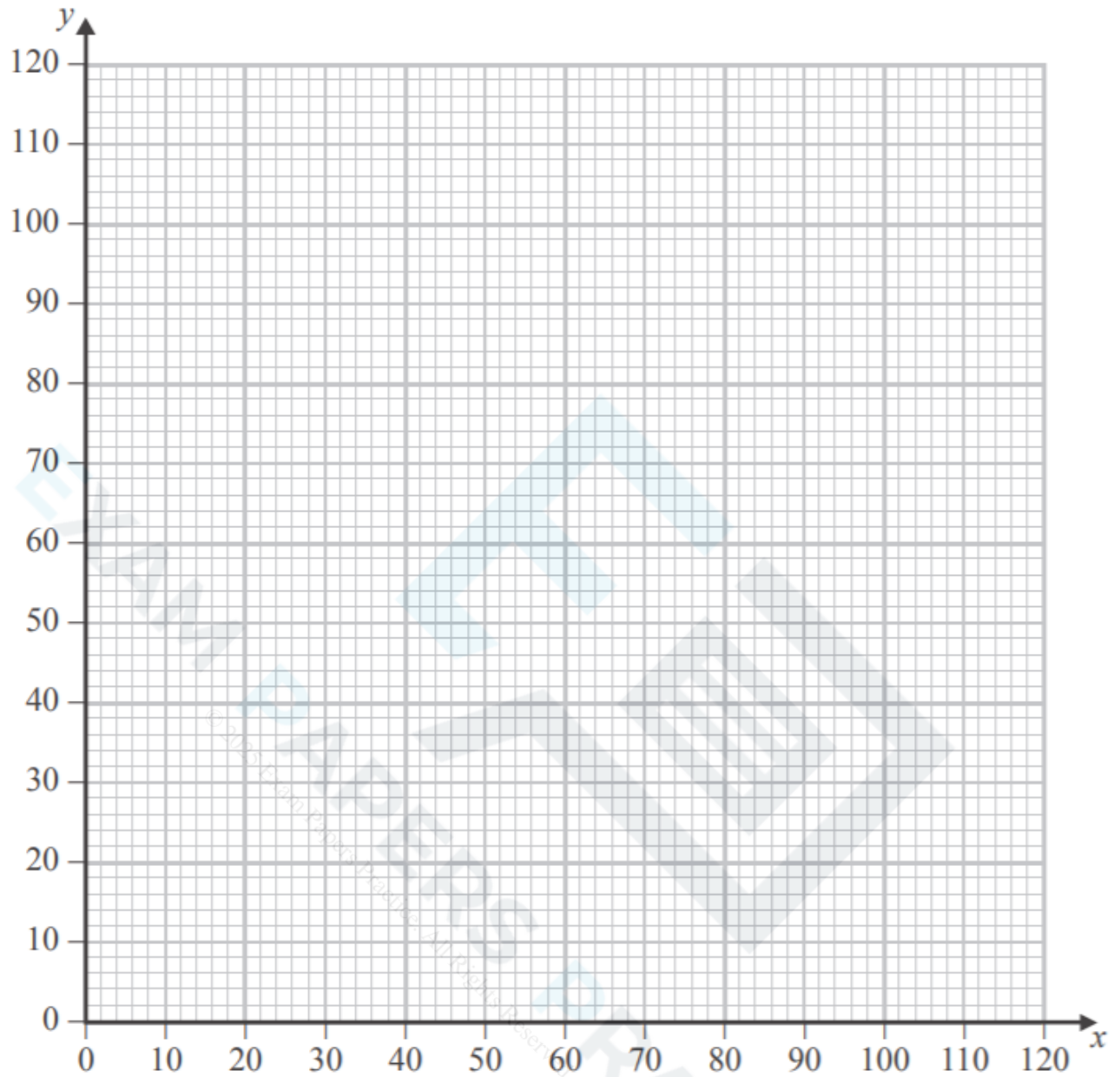


Diagram 1